



C++ as better C-II

1. Function overloading
2. const
3. Inline functions

- Need with different programming scenarios
 - Mathematical operations like add, subtract etc
 - Logical operations like Max, min, Average
- Multiple functions with same names but different signature
- Signature of a function:
 - Data types of parameters
 - Number of parameters
 - Sequence of parameters
 - Return type is NOT part of signature
- Compiler selects best-fit function.
 - Ambiguity error with examples
- Name mangling
 - Names are changed by compiler
 - Compiler specific
- Examples of correct and in-correct overloaded functions

- Need with different programming scenarios
 - Mathematical constants like pi, g etc
- const variable
- const object and member function
- const pointer to data
- pointer to const data

- Function call overheads
- Inline function to avoid these overheads
- Creating inline functions
 - Using inline keyword
 - By defining function inside the class itself
 - Even global functions can be inline
- Inline is a request. May get rejected
- Benefits of inline over macros
 - Better type checking
 - Can access private data of class